JOHN PAUL MORENO

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EDUCATION

The University of Texas at Austin | Bachelor of Science in Mechanical Engineering | GPA: 3.7 May 2028

Awards: GeoFORCE STEM Four Year Scholarship ($60,000), HSF Scholar (2024, 2025), Dell Scholar (2024, 2025), Ramshorn Engineering Scholar (2024, 2025) AP Scholar (2024), Hispanic Recognition Award (SAT)

SKILLS

CAD (SolidWorks, Fusion360), 3D Printing, Laser Cutting (Trotec), Prototyping, Report Writing, Teamwork

Analysis: FEA, MATLAB

Certifications: Walker Department of Mechanical Engineering Shop Certification

TECHNICAL EXPERIENCE

Gel Blaster Nexus Aug. 2026 - Present

Mechanical Engineering Intern

- Led the prototyping and design for manufacturing in Fusion360 for 3D printed PCB housings using ABS.
- Refined packaging of electronic wiring for 4 different PCB types and routed 20+ ft in wiring into 10 x 8 in housing.
- Reduced material costs for Battle Bay product assembly by 50%, redesigning frame geometry validated by FEA.

Mirada3D – Equipment 3D Models Nov. 2025 – May 2026

CAD Modeling Engineer

- Converted 2D engineering drawings into scaled down parametric industrial-equipment showcase assemblies in Fusion 360.
- Produced over 100+ detailed sketches for 3D printed components by refining geometry down to .01 in tolerances.
- Fabricated custom box display cases with 6 laser-cut acrylic panels to present deliverables and meet project deadlines.

Undergraduate Research Jan. 2025 – Apr. 2025

High-Temperature Conductivity of Composite Materials

- Reconfigured a four-point probe device in SOLIDWORKS, incorporating torsion spring components and PTFE material selection research.
- Conducted in-house testing at temperatures above 250°C with 1,000+ resistance/voltage data points to analyze conductivity trends.
- Machined PTFE block using manual mill to adapt fasteners and springs; worked with insulating resin compound to adhere components.

ME 302 Battle Bot (1-lb) Jan. 2025 – Apr.2025

- Drafted main design for a 1-lb battle bot, integrating 3 electric motors, three 20A ESCs, servo motor, kill switch and 7.4V battery pack.
- Designed, FEA-validated, and PLA 3D-printed custom wheels with interference fit and fastener access to integrate with N20 DC motors.
- Laser-cut a toleranced protective lid out of 3mm plywood for easy access to electronics while adhering to weight constraints.

LEADERSHIP EXPERIENCE

Longhorn Baja (Society of Automotive Engineers) Sept. 2025 – Present

Racing Team – Ergonomics Lead

The University of Texas at Austin's collegiate design competition team that designs and optimizes a single-seat off-road vehicle.

- Led pedal box geometry and prototyping, after evaluation pivoted to premanufactured pedals to reduce risk.
- Optimized aluminum body-panels by reducing sheet sizes from 4x10 ft to 3x4 ft, lowering material costs by over $600.
- Coordinated modification of racing seat mounting plate to integrate to chassis; routed brake lines; finalized driver harness mounting point.

Society of Hispanic Professional Engineers (SHPE) Aug. 2024 – Present

- Mentor in SHPE for Big & Little program, advising student in mechanical engineering coursework and technical opportunities.
- Community outreach volunteer in SHPE Jr. to inform and guide 40+ students at Del Valle High School transition from high school to college.